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ORIGINAL ARTICLE

Risk aversion and the predictability of crude oil market volatility: A forecasting experiment with random forests

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ABSTRACT

We analyze the predictive power of time-varying risk aversion for the realized volatility of crude oil returns based on high-frequency data. Using random forests, and their extensions to quantile random forests and extreme random forests, we show that risk aversion improves out-of-sample accuracy of realized volatility forecasts. The predictive power of risk aversion is robust to various covariates including realized skewness and realized kurtosis, various measures of jump intensity, and leverage. Our findings highlight the importance of non-cash flow factors over commodity-market uncertainty with significant implications for the pricing and forecasting in these markets.

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1. Introduction

The financialization of commodity markets has resulted in increased participation of hedge funds, pension funds, and insurance companies in oil-based financial products (e.g. Hamilton & Wu, 2015). While several researchers argue that supply and demand fundamentals serve as the main drivers of commodity prices (e.g. Irwin & Sanders, 2011), others argue that speculative activities play a significant role in risk premiums and volatility (e.g. Singleton, 2014). Indeed, recent evidence suggests that investors' risk preferences towards the oil price contain a business-cycle component (Christoffersen & Pan, 2017) and that risk premiums embedded in crude oil prices are determined by the risk aversion of market participants (Li, 2018). Further motivated by the evidence in Robe and Wallen (2016) that implied volatilities extracted from crude oil options are driven partly by the VIX, considered as a proxy for risk aversion and macro uncertainty, we provide novel insight into the predictive value of risk preferences for crude oil-returns volatility by utilizing the measure of time-varying risk aversion recently developed by Bekaert et al. (2017). By doing so, we contribute to the literature on the predictive power of behavioral factors over commodity-market dynamics.

Considering that crude oil is arguably the world's most strategic commodity, accurate forecasts of oil-return volatility are of paramount importance to not only oil traders and corporate decision-makers, but also to policymakers as oil market uncertainty can negatively impact economic activity and contribute to aggregate macroeconomic uncertainty (Aye et al., 2014; van Eyden et al., 2019). Naturally, a large literature exists on the forecastability of oil-price-return volatility using various types of Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models (e.g. Chkili et al., 2014; Efimova & Serletis, 2014) with further extensions based on the Markov-switching multifractal (MSM) model (e.g. Lux et al., 2016). These studies, however, generally examine daily oil returns without taking advantage of the information contained in intraday dynamics. In fact, numerous works including Andersen et al. (2003, 2010) and Hansen and Lunde (2005) suggest that the rich information contained in intraday data can be utilized to produce more accurate estimates and forecasts of daily return volatility. In light of this, several researchers including Degiannakis and Filis (2017), Chen et al. (2019), and Gkillas et al. (2020) use variants of the Heterogeneous Autoregressive (HAR) model developed by Corsi (2008) to forecast the realized volatility (RV) of crude oil returns (i.e. the sum of non-overlapping squared high-frequency oil returns observed within a day).[1] The popularity of the HAR-RV model stems from its ability to capture important stylized facts of volatility such as long memory and multi-scaling behavior. In fact, barring

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the recent studies of Degiannakis and Filis (2017), Chen et al. (2019) and Gkillas et al. (forthcoming), earlier works based on intraday data generally conclude that alternative models are not able to outperform the forecasting accuracy of a simple HAR-RV model. In contrast, Degiannakis and Filis (2017) show that it is possible to beat the HAR-RV model by incorporating information on the volatilities of four different asset classes (stocks, currencies, commodities, macroeconomic policy), while Gkillas et al. (forthcoming) show that extending the baseline linear HAR-RV model to incorporate an index of financial stress improves forecast accuracy once they account for the asymmetry of a forecaster's loss function. At the same time, Chen et al. (2019) show that allowing for time-variation and accounting for asymmetric jumps and co-jumps with the equity (S&P 500) market can outperform the benchmark HAR-RV model.

Against this backdrop, we extend the literature on forecasting realized oil-returns volatility (derived from intraday data) by incorporating a measure of time-varying risk aversion developed by Bekaert et al. (2017) into the HAR-RV framework. This measure of risk-aversion distinguishes the time variation in economic uncertainty (the amount of risk) from time variation in risk aversion (the price of risk) and provides an unbiased representation of time-varying risk aversion in financial markets.[2] From an economic perspective, the predictive value of risk aversion over crude oil-returns volatility stems from at least two distinct channels. Considering the evidence that investor sentiment has forecasting power over future macroeconomic conditions (e.g. Ludvigson, 2004) and that real economic activity drives oil prices (e.g. Kilian & Hicks, 2013), real economic activity is one possible channel.[3] A second channel operates through the speculative activities in the marketplace. Li (2018) notes that crude oil-risk premiums are low when the level of speculation is high (that is when aggregate risk aversion of market participants is low). Therefore, it is possible that changes in risk aversion serve as a determinant of speculative behavior in the oil market and, thereby, capture predictive information over volatility. To the best of our knowledge, ours is the first study to utilize the time-varying measure of risk aversion developed by Bekaert et al. (2017) for forecasting crude oil-returns volatility.

On the technical side, we use variants of random forests to model the realized volatility of crude oil returns. Random forests capture possible nonlinear links between realized volatility and risk aversion in a data-driven way and allow us to study the predictive value of risk aversion in the presence of other covariates including upward and downward realized volatility, realized skewness and kurtosis, and jumps (see, e.g. Lyócsa & Molnár, 2016; Mei et al., 2017). While random forests are only one among a large number of machine-learning techniques (neural networks, support vector machines, etc.), the decision to use random forests for our empirical analysis stems from the fact that random forests are a simple integrated modeling platform that allows potentially complex links between realized volatility and its predictors to be modeled in a data-driven way. Moreover, various variants of random forests (quantile random forests, extreme random forests; Section 4.3) are available, allowing us to assess the robustness of our results without having to introduce completely different machine-learning techniques. Although random forests, including quantile random forests, and competing forecasting models like neural networks and particle swarm optimization, are utilized for forecasting the volatility of commodity-price returns (e.g. Pradeepkumar & Ravi, 2017), the literature offers limited evidence on the forecastability of realized volatility computed from intraday data and the role of risk aversion as a predictor of volatility has not been examined in this context. Looking forward, our forecasting results show that random forests, including their extensions to quantile random forests and extreme random forests, recover that risk aversion improves out-of-sample accuracy of realized volatility forecasts, highlighting the predictive information captured by non-cash flow related factors over commodity market fluctuations.

We proceed as follows. We describe our empirical methods in Section 2 and our data in Section 3. We summarize our empirical results in Section 4 and conclude in Section 5.

2. Methods

2.1. The HAR-RV model

Like Andersen et al. (2012), we measure daily realized volatility (RV) by the median realized variance (MRV) computed from intraday data as

$$MRV_t = \frac{\pi}{6 - 4\sqrt{3} + \pi} \frac{T}{T-2} \sum_{i=2}^{T-1} \text{med}\ (|r_{t,i-1}|, |r_{t,i}|, |r_{t,i+1}|)^2, \quad (1)$$

where $r_{t,i}$ = intraday return i within day t, and $i = 1, .., T$ denotes the number of intraday observations. The estimator, MRV, is considered to be robust to jumps and less biased in the presence of market-microstructure noise.[4]

The baseline model to forecast realized volatility follows the widely-studied HAR-RV model (Corsi, 2008) described as (h = forecast horizon)

$$RV_{t+h} = \beta_0 + \beta_d RV_t + \beta_w RV_{w,t} + \beta_m RV_{m,t} + \epsilon_{t+h}, \tag{2}$$

where $RV_{t+h} = \text{mean}\{RVt+1, ..., RVt+h\}$, $RV_{w,t}$ = average RV from day $t-5$ to day $t-1$, and $RV_{m,t}$ = average RV from day $t-22$ to day $t-1$. We analyze a short daily forecast horizon ($h=1$), a medium weekly forecast horizon ($h=5$), and a long monthly forecast horizon ($h=22$).[5]

The HAR-RV model uses volatilities from different time resolutions to forecast realized volatility and, thereby, formalizes the heterogeneous market hypothesis (Müller et al., 1997). In our context, this hypothesis translates into a setting in which the market for oil is populated by different groups of market participants, with short- and long-term traders responding to information flows at different time horizons. In this regard, the HAR-RV model accounts for the presence of different groups of agents including managed money traders, who are subject to funding constraints (Acharya et al., 2013), and commodity index traders (Hamilton & Wu, 2014), who may be less focused on short-term interactions.

We use the HAR-RV model as a nucleus for building more complex models. One such model is the HAR-RV-RA model, which includes risk aversion, RA_t:

$$RV_{t+h} = \beta_0 + \beta_d RV_t + \beta_w RV_{w,t} + \beta_m RV_{m,t} + \theta RA_t + \epsilon_{t+h}. \tag{3}$$

2.2. The HAR-RV-RF model

Following the notation used by Hastie et al. (2009, Chapter 9), a regression tree, T, that features J terminal nodes recursively partitions the space of covariates, $\mathbf{x} = (x_1, x_2, ...)$, into l non-overlapping regions, R_l. The list of covariates comprises the standard HAR-RV predictors (that is, $RV_t, RV_{w,t}$, and $RV_{m,t}$), risk aversion, and several other covariates. Regression trees render it possible to study an arbitrarily large number of covariates via a greedy algorithm that determines the partitions in a top-down and binary way. At the top level of a regression tree, the algorithm chooses the first partition in such a way that the partitioning covariate, s, and its partitioning point, p, define the half-planes $R_1(s,p) = \{x_s | x_s \leq p\}$ and $R_2(s,p) = \{x_s | x_s > p\}$ that minimize the loss function:

$$\min_{s,p} \left\{ \min_{\overline{RV}_1} \sum_{x_s \in R_1(s,p)} (RV_i - \overline{RV}_1)^2 + \min_{\overline{RV}_2} \sum_{x_s \in R_2(s,p)} (RV_i - \overline{RV}_2)^2 \right\}, \tag{4}$$

where i denotes the $t+h$ data on realized volatility that belong to the half-planes, $\overline{RV}_k = \text{mean}\{RV_i | x_s \in R_k(s,p)\}, k = 1,2$ denotes the half-plane-specific mean of realized volatility, and the inner minimization chooses the half-plane-specific means to minimize the region-specific squared error loss.[6] The outer minimization searches over all possible combinations of s and p and gives the first optimal partitioning predictor and the first optimal partitioning point and, thereby, a new regression tree with two terminal nodes.

In the next step, the minimization setup described in Equation (4) is carried on to this new regression tree, now separately for the two optimal top-level half-planes. The minimization yields up to two second-level optimal partitioning covariates, two second-level optimal partitioning points, and four second-level region-specific means. This minimize-and-partition search process is repeated until the regression tree has a preset maximum number of terminal nodes or every terminal node has a minimum number of observations.[7] The final regression tree sends the covariates into the R_l optimal regions and then predicts realized volatility by its region-specific mean ($\mathbf{1}$ = indicator function):

$$T\left(\mathbf{x}_i, \{R_l\}_1^L\right) = \sum_{l=1}^{L} \overline{RV}_l \mathbf{1}(\mathbf{x}_i \in R_l). \tag{5}$$

Fine predictions of realized volatility can be computed by growing a large regression tree. However, because a large regression tree is data-sensitive due to its hierarchical structure, a more efficient way to compute predictions is to combine a large number of regression trees. One approach used in this regard is a bootstrap aggregation (bagging, Breiman, 1996): 1) A large number of bootstrap samples are obtained from the data. 2) A regression tree is estimated on every sample. 3) Realized volatility is predicted by averaging over the bootstrapped regression trees. An alternative approach, the one we adopt in this research, is to grow random forests (Breiman, 2001). Growing a random forest resembles bagging in that it requires drawing a large number of samples from the data. In contrast to bagging, a random regression tree is estimated on every sample. A random regression tree uses for every partitioning step a random subset of the covariates. Random forests, thereby, mitigate the influence of individual covariates, decorrelate the predictions from individual trees, and stabilize the predictions of realized volatility.[8]

2.3. Other covariates of realized volatility

Earlier studies have used several covariates of realized volatility to extend the standard HAR-RV model. Important covariates are (i) various types of jumps (i.e. downside (bad) jumps, upside (good) jumps, asymmetric jumps, truncated large jumps,

truncated small jumps), and, (ii) realized skewness and kurtosis. Our methodology for the detection of jumps is based on a benchmark scheme developed by Huang and Tauchen (2005) and Barndorff-Nielsen and Shephard (2005).

2.3.1. Detecting Jumps in the price process

The procedure to detect jumps is based on the decomposition of realized volatility into a jump and a continuous component.[9] This procedure requires an estimator that excludes jumps. Barndorff-Nielsen and Shephard (2005) propose an accurate estimate of the integrated variance excluding jumps, named bipower variation (BPV), formulated as:

$$BPV_t = \xi_1^{-2} \sum_{i=2}^{T} |r_{t,i}||r_{t,i-1}|, \tag{6}$$

where $\xi_1 = \sqrt{(2/\pi)} = E(|Z|)$ denotes the absolute value of the mean of a standard normally distributed random variable, Z. As in Andersen et al. (2007), total jumps are detected by the ratio-statistic $(ZJ_t^{(BPV)})$ given by

$$ZJ_t^{(BPV)} = \frac{(RV_t - BPV_t)RV_t^{-1}}{\left(\left(\xi_1^{-4} + 2\xi_1^{-2} - 5\right) max\left\{1, TQ_t BPV_t^{-2}\right\}\right)^{1/2}} \rightarrow N(0,1), \tag{7}$$

where $RV_t = \sum_{i=1}^{T} r_{t,i}^2$ is the realized variance defined as a consistent estimator of the integrated variance plus the jump contribution, and the realized tripower quarticity is given by $TQ_t = T\xi_{4/3}^{-3} \sum_{i=3}^{T} |r_{t,i}|^{4/3}|r_{t,i-1}|^{4/3}|r_{t,i-2}|^{4/3}$. TQ_t is an asymptotically unbiased estimator of integrated quarticity in the absence of microstructure noise and converges in probability to integrated quarticity. The $ZJ_t^{(BPV)}$ is useful as a pre-test, prior to detecting jumps under the null hypothesis of no jumps. Finally, total jumps are detected by applying a jump-detection scheme:

$$RJ_t = \mathbf{1}_{\left[ZJ_t^{(BPV)} > \Phi_a\right]} |RV_t - BPV_t|, \tag{8}$$

where $\mathbf{1}$ = indicator function for $ZJ_t^{(BPV)}$ exceeding a given critical value. A jump occurs when $ZJ_t^{(BPV)}$ exceeds the critical value of the normal distribution, denoted by Φ_a, at a a level of significance.

We rely on this jump-detection technique to recover upside/downside, asymmetric, and truncated large/small jumps. Note that even though the jump component is a random variable (that captures the random arrival of new information), past jumps are observable. Besides, it is expected jumps that may have predictive value for realized volatility (as two random variables can be correlated at different leads and lags). The main reason is that jumps occur during periods of high volatility due to large price movements in the market. From a statistical point of view, large price movements may appear in clusters, and hence, jumps may help to predict future price dynamics.

2.3.2. Upside, downside, and asymmetric jumps

Good and bad realized volatility (semi-variance) can serve as measures of downside and upside risk and capture the sign asymmetry in the price process. Chen et al. (2019) use HAR-type models along with realized semivariances and signed jumps as in Patton and Sheppard (2015) to forecast crude oil volatility. In line with Barndorff-Nielsen et al. (2010), we compute bad and good realized semivariance (RV^B and RV^G) as (the "simple" realized volatility is given by $RV^S = RV^B + RV^G$):

$$RV_t^B = \sum_{i=1}^{T} r_{t,i}^2 \mathbf{1}_{\left[(r_{t,i})<0\right]}, \tag{9}$$

$$RV_t^G = \sum_{i=1}^{T} r_{t,i}^2 \mathbf{1}_{\left[(r_{t,i})>0\right]}. \tag{10}$$

Like Duong and Swanson (2011, 2015), we estimate downside (bad), upside (good) jumps as:

$$RJ_t^- = \mathbf{1}_{\left[ZJ_t^{(BPV)} > \Phi_a\right]} \sum_{i=1}^{T} |r_{t,i}|^q \mathbf{1}_{\left[(r_{t,i})<0\right]}, \tag{11}$$

$$RJ_t^+ = \mathbf{1}_{\left[ZJ_t^{(BPV)} > \Phi_a\right]} \sum_{i=1}^{T} |r_{t,i}|^q \mathbf{1}_{\left[(r_{t,i})>0\right]}, \tag{12}$$

where q = asymmetry parameter that affects the limiting behavior of the estimator. We are interested in the case where $q \geq 2$ (large values of q, are dominated by large jumps).[10] We set $q = 2.5$, as Duong and Swanson (2011) find more significant HAR coefficients for a range of q from 2 to 6.

Finally, we compute the asymmetric-jump component as:

$$RJA_t = \mathbf{1}_{\left[ZJ_t^{(BPV)} > \Phi_a\right]} \left(RJ_t^+ - RJ_t^-\right). \tag{13}$$

A negative (positive) sign implies that the jump occurred on day t is associated with events that primarily affected the bad (good) volatility.

2.3.3. Truncated large and small jumps

In line with Duong and Swanson (2011), we estimate truncated large ($RVLJ_t$) and small jumps ($RVSJ_t$) using a decomposition based on a fixed truncation level (γ) (a threshold method to select the appropriate value of the parameter, γ, that can separate $RVLJ_t$ from $RVSJ_t$). Noting the limited literature on forecasting oil volatility, particularly taking into account the forecasting ability of $RVLJ_t$ and $RVSJ_t$, Liu et al. (2018) show that HAR-type models

Table 1. Full-sample estimates (HAR-RV model).

	$h=1$					
Model	Intercept	MRV	MRV_w	MRV_m	RA	Adj. R2
HAR-RV	3.2286	10.0487	5.2102	4.8466	–	0.7364
p-value	0.0012	0.0000	0.0001°	0.0001°	–	–
HAR-RV-RA	−1.9150	9.3804	5.3565	4.7750	2.9494	0.7389
p-value	0.0555	0.0001°	0.0001°	0.0001°	0.0032	–
	$h=5$					
Model	Intercept	MRV	MRV_w	MRV_m	RA	Adj. R2
HAR-RV	2.4619	10.9463	3.7549	5.0119	–	0.7876
p-value	0.0138	0.0001°	0.0002	0.0001°	–	–
HAR-RV-RA	−1.7560	10.4366	3.7681	4.8818	2.6756	0.7919
p-value	0.0791	0.0001°	0.0002	0.0001°	0.0075	–
	$h=22$					
Model	Intercept	MRV	MRV_w	MRV_m	RA	Adj. R2
HAR-RV	3.1752	9.3592	2.2664	2.9516	–	0.7386
p-value	0.0015	0.0001°	0.0234	0.0032	–	–
HAR-RV-RA	−1.8773	8.9473	2.8770	3.1814	5.2033	0.7547
p-value	0.0605	0.0001°	0.0040	0.0015	0.0001°	–

Note: Estimated coefficients are scaled by their estimated standard error. RA = Risk aversion. p-values are based on Newey-West standard errors. ° denotes a value smaller than 0.0001. Adj. R2 = adjusted coefficient of determination.

including both $RVLJ_t$ and $RVSJ_t$ can provide superior forecasting performance in case of oil futures data.[11] We compute $RVLJ_t$ and $RVSJ_t$ as:

$$RVLJ_t = min\left\{RJ_t, \mathbf{1}_{\left[ZJ_t^{(BPV)}>\Phi_a\right]}\left(\sum_{i=1}^{T} r_{t,i}^2 \mathbf{1}_{|r_{t,i}|\geq\gamma}\right)\right\}, \tag{14}$$

$$RVSJ_t = RJ_t - RVLJ_t, \tag{15}$$

where we set $\gamma = 2$ in order to capture large size jumps (Duong & Swanson, 2011).

2.3.4. Realized skewness and kurtosis

Following Amaya et al. (2015), we consider realized skewness, *RSK*, as a measure of the asymmetry of the daily returns distribution and realized kurtosis, *RKU*, as a measure of extreme occurrences. We compute RSK_t and RKU_t as

$$RSK_t = \frac{\sqrt{T}\sum_{i=1}^{T} r_{t,i}^3}{\left(\sum_{i=1}^{T} r_{t,i}^2\right)^{3/2}}, \tag{16}$$

$$RKU_t = \frac{T\sum_{i=1}^{T} r_{t,i}^4}{\left(\sum_{i=1}^{T} r_{t,i}^2\right)^{2}}. \tag{17}$$

Scaling by $\sqrt{T}$ (T) ensures that the statistic reflects daily skewness (kurtosis).

3. Data

We use intraday data on West Texas Intermediate (WTI) oil futures that are traded on NYMEX over a 24 h trading day (pit and electronic) to construct daily measures of realized volatility, the corresponding good, and bad variants, and the other covariates described in Section 2.3. The futures price data, in continuous format, are obtained from www.disktrading.com and www.kibot.com. Close to the expiration of a contract, the position is rolled over to the next available contract, provided that activity has increased. Daily returns are computed as the end of the day (New York time) price difference (close to close). In the case of intraday returns, 1-minute prices are obtained via last-tick interpolation (if the price is not available at the 1-minute stamp, the previously available price is imputed), and 5-minute returns are then computed by taking the log-differences of these prices.[12]

Table 2. Out-of-sample results (HAR-RV model).

Loss function	$h=1$	$h=5$	$h=22$
*L*1	0.8096	0.6409	0.4526
*L*2	0.6228	0.6437	0.0133

Note: p-values of Diebold-Mariano tests (HAR-RV vs. HAR-RV-RA forecasts) for three different forecast horizons. Null hypothesis: the series of forecasts are equally accurate. Alternative hypothesis: the forecasts from the HAR-RV model that features the risk aversion are more accurate. *L*1: absolute loss. *L*2: quadratic loss. The models were estimated using a rolling-estimation window (2350 observations).

As a measure of risk aversion, we utilize the risk-aversion index of Bekaert et al. (2017), which is available for download from https://www.nancyxu.net/risk-aversion-index. This measure of time-varying risk aversion is calculated from observable financial information at high (daily) frequencies and relies on a set of six financial instruments (term spread, credit spread, de-trended dividend yield, realized and risk-neutral equity return variance, realized corporate bond return variance). An important feature of this measure is that it distinguishes time variation in economic uncertainty (the amount of risk) from time variation in risk aversion (the price of risk). Based on the availability of the risk-aversion data, the sample period is February 1997 to December 2016.

4. Empirical findings

4.1. Results for the standard HAR-RV model

Table 1 presents the full-sample estimation results for the HAR-RV and HAR-RV-RA models.[13] The findings show that the HAR-RV model generally fits the data well with statistically significant estimated coefficients at all three forecasting horizons, where the minimum value of the adjusted coefficient of determination is 0.7364. As for the HAR-RV-RA model, the coefficient estimated for risk aversion is significant and indicates that a higher level of risk aversion predicts a subsequent increase in realized volatility. Christoffersen and Pan (2017) note that an increase in economic activity lowers investors' relative risk aversion towards oil-price changes and large oil-price fluctuations are associated with deteriorating economic conditions. Accordingly, the observed positive predictive value of risk aversion

Table 3. Out-of-sample results (random forests).

Loss function	$h=1$	$h=5$	$h=22$
	Max. no. of terminal nodes = 5		
$L1$	0.0000	0.0000	0.0115
$L2$	0.0002	0.0144	0.1230
	Max. no. of terminal nodes = 10		
$L1$	0.0000	0.0000	0.0475
$L2$	0.0116	0.0253	0.0936
	Min. terminal node size = 10		
$L1$	0.0010	0.0367	0.3717
$L2$	0.0040	0.0480	0.1417

Note: p-values of Diebold-Mariano tests (HAR-RV-RF vs. HAR-RV-RF-RA forecasts) for three different forecast horizons. Null hypothesis: the series of forecasts are equally accurate. Alternative hypothesis: the forecasts from the HAR-RV-RF model that features the risk aversion are more accurate. $L1$: absolute loss. $L2$: quadratic loss. The models were estimated using random forests and a rolling-estimation window (2350 observations).

Table 4. Ranking of covariates.

Covariate	$h=1$	$h=5$	$h=22$
MRV	1.83	2.20	3.13
MRV_w	2.65	1.82	1.19
MRV_m	4.94	3.88	3.09
Asymmetric jumps	13.33	13.85	13.50
Bad jumps	9.71	10.12	10.20
Good jumps	9.92	9.13	9.06
Large jumps	7.08	6.95	7.18
Small jumps	8.63	8.61	9.14
Returns	9.05	9.73	9.97
Bad realized volatility	2.00	2.91	2.98
Good realized volatility	3.81	4.19	4.73
Realized skewness	12.99	13.32	13.40
Realized kurtosis	12.84	12.28	11.96
Leverage	14.62	14.33	14.56
Risk aversion	6.61	6.67	5.90

Note: The mean rank of a covariate across all rolling-estimation windows (2350 observations) is computed based on its effect on node purity as measured by the residual sum of squares. The number of trees per random forest is 500, and the maximum number of terminal nodes is five.

over volatility may be driven by the real economic activity channel and possible comovement of risk aversion with changes in economic conditions. We also observe that the adjusted coefficient of determination increases only slightly for short forecast horizons when we add risk aversion to the model, while risk aversion makes a somewhat larger contribution to the overall fit of the model in the case of the medium and long forecast horizons.

We report in Table 2 the p-values of Diebold and Mariano (1995) tests to examine whether risk aversion helps to improve out-of-sample forecast accuracy ($L1$ = loss function depends on the absolute forecast error, $L2$ = loss function depends on the squared forecast error). To this end, we compute out-of-sample forecast errors for the three forecast horizons by means of a rolling-estimation window of length 2,350 observations, which corresponds to half of the sample size. We use a rolling-estimation window because the realized volatility of oil-price returns may be subject to structural breaks (Wen et al., 2016). We then derive the test results using the well-known modified Diebold-Mariano test proposed by Harvey et al. (1997).[14] The test results suggest that risk aversion hardly helps to improve out-of-sample forecast accuracy.[15]

4.2. Results for the HAR-RV-RF model

Considering that the level of speculation is negatively correlated with the aggregate risk aversion of market participants (Li, 2018) and that speculators require commodity positions in their portfolios for diversification purposes, particularly during periods of uncertainty (Hamilton & Wu, 2014), the informational content of high levels of risk aversion during times of crisis for subsequent realized volatility may differ from the informational content of low levels of risk aversion observed during normal times.[16] As stated earlier, random forests are a natural modeling platform to capture such a nonlinear data structure, and they allow us to control for the impact of several additional covariates other than risk aversion. Following the previous works on realized volatility in different contexts, we use the following covariates in our forecasting model in addition to risk aversion (and the standard HAR-RV predictors): measures of good and bad volatility, realized skewness and realized kurtosis, oil returns, a leverage effect defined as $\max(\text{returns}_t, 0)$, and various jump measures including asymmetric jumps, good/bad jumps and small/large jumps.

We estimate random forests using the full list of covariates, with and without risk aversion, again using a rolling-estimation window length of 2,350 observations.[17] This procedure yields two time series of forecasts (one for the HAR-RV-RF and one for the HAR-RV-RF-RA model). We consider three different specifications (maximum number of terminal tree nodes = 5 (10); minimum node size = 10). The results of Diebold-Mariano tests (Table 3) are similar across the three specifications and support the hypothesis that time-varying risk aversion contains significant predictive value for realized volatility. The tests yield significant results in the case of the short and medium forecast horizon. For the long forecast horizon two out of three tests are significant under $L1$ loss.

Table 4 shows a ranking of the covariates according to their relative importance.[18] Daily realized volatility and its weekly and monthly counterparts (standard HAR-RV predictors) are among the top-ranked predictors, followed by realized skewness and realized kurtosis. Risk aversion has an average rank of about six, indicating its superior predictive ability compared to returns and leverage and, importantly, the various jump covariates.[19]

In Table 5, we summarize the pseudo-R-squared of the HAR-RV-RF models (averaged across rolling-estimation windows).[20] The pseudo-R-squared of the HAR-RV-RF models is somewhat larger for the

Table 5. Pseudo R-squared (HAR-RV-RF-RA model).

Specification	$h=1$	$h=5$	$h=22$
Max. no. of terminal nodes = 5	0.7468	0.7953	0.7636
Max. no. of terminal nodes = 10	0.7663	0.8183	0.7902
Min. terminal node size = 10	0.7737	0.8356	0.8302

Note: The pseudo-R-squared is defined as $1-MSE_{oob}/Var$, where MSE_{oob} denotes the mean-squared error that obtains when the estimated model is used to compute the out-of-bag data (that is, the data not used for tree building when bootstrapping a random forest) and *Var* is an estimate of the variance of the dependent variable. The number of trees was set to 500.

medium and long forecast horizons than for the short forecast horizon. In all cases, the pseudo-R-squared values exceed the full-sample adjusted R-squared values for the HAR-RV-RA models (Table 1), especially when we fix the minimum terminal node size at 10 observations.

In Table 6, we vary the length of the rolling-estimation window in increments of 100 observations, starting with 1800 up to 2800. Our findings are robust to the size of the rolling-estimation window, suggesting that risk aversion carries significant predictive information. In addition, we report in Table 7 the results for the test proposed by Clark and West (2007). All tests yield significant results, corroborating the results of the Diebold-Mariano tests.

From an economic perspective, the predictive value of time-varying risk aversion supports the finding by Christoffersen and Pan (2017) that relates investors' risk aversion to oil-price fluctuations as changes in risk preferences reflect changes in economic conditions. Accordingly, it can be argued that the observed positive predictive value of risk aversion over oil market volatility is at least partially driven by the real-economic-activity channel and the comovement of risk aversion with changes in economic conditions. In fact, the positive predictive relationship between risk aversion and oil market volatility is in line with the finding in Robe and Wallen (2016) that relates implied volatility in crude oil options to risk aversion and macroeconomic uncertainty. Given the close association between oil prices and real economic activity (Kilian & Hicks, 2013), our findings support the presence of a business-cycle component embedded in oil prices, which is predictable by non-cash flow factors.

4.3. Several extensions

As a first extension, we use quantile random forests to estimate HAR-RV-QRF models to study the quantiles of the conditional distribution of realized volatility. A quantile random forest, proposed by Meinshausen (2006) as an extension of random forests, does not only store the information on the mean of realized volatility at the leaves but rather keeps all observations of realized volatility in order

Table 6. Out-of-sample results (alternative rolling-window lengths).

Window	$h=1$	$h=5$	$h=22$
1800	0.0000	0.0176	0.1120
1900	0.0049	0.0155	0.0883
2000	0.0014	0.0054	0.0911
2100	0.0000	0.0086	0.0878
2200	0.0001	0.0058	0.0965
2300	0.0000	0.0136	0.1155
2400	0.0005	0.0158	0.1198
2500	0.0001	0.0077	0.1227
2600	0.0001	0.0125	0.1044
2700	0.0277	0.1717	0.3586
2800	0.3130	0.2305	0.0167

Note: p-values of Diebold-Mariano tests (HAR-RV-RF vs. HAR-RV-RF-RA forecasts) for three different forecast horizons. Null hypothesis: the series of forecasts are equally accurate. Alternative hypothesis: the forecasts from the HAR-RV-RF model that features the risk aversion are more accurate. Loss function: *L*2. The models were estimated using random forests consisting of 500 trees and a maximum of five terminal nodes. Window = length of the rolling-estimation window.

Table 7. Clark-west-tests (HAR-RV-RF model).

Specification	$h=1$	$h=5$	$h=22$
Max. no. of terminal nodes = 5	2.6076	1.9376	1.6034
Max. no. of terminal nodes = 10	1.8527	1.7524	1.7491
Min. terminal node size = 10	2.6019	2.1078	1.8252

Note: t-statistics of the Clark-West tests are based on Newey-West standard errors. The number of trees was set to 500. Critical values (one-sided test) are: 1.282 (10%) and 1.645 (5%).

to subsequently compute an estimate of its conditional distribution function. This distribution function is then used to inspect the incremental predictive value of risk aversion for realized volatility at the quantile level.[21]

Table 8 summarizes the results of Diebold-Mariano tests for the HAR-RV-QRF model, where we use the standard check function underlying analyses of quantile regressions to evaluate the forecast errors. In other words, we explicitly take into account the asymmetry of a forecaster's loss function based on the assumption that a forecaster targets quantiles of the conditional distribution of realized volatility rather than the median.[22] The findings are generally in line with those obtained from the standard HAR-RV-RF model in that risk aversion helps to improve forecast accuracy.

As a second extension, we consider extreme random forests as developed by Geurts et al. (2006).[23] Extreme random forests expand the advantages of standard random forecasts over bootstrap aggregation in that, given the list of randomly selected covariates that can be used for splitting at a node, a (uniform) random split is sampled for every covariate in this list (using the full sample rather than a bootstrap sample), and then the best split/covariate combination is selected among these randomly sampled splits. The test results for the HAR-RV-ERF model (Table 9) are significant for the short and medium forecast horizon and further corroborate that risk aversion captures predictive information over the path of future realized volatility.

Table 8. Out-of-sample results (HAR-RV-QRF model).

Loss function	*L*1			*L*2		
Quantile	$h=1$	$h=5$	$h=22$	$h=1$	$h=5$	$h=22$
0.02	0.0000	0.0000	0.0003	0.0000	0.0001	0.0273
0.1	0.0000	0.0000	0.0000	0.0000	0.0003	0.0117
0.2	0.0000	0.0000	0.0000	0.0000	0.0103	0.0433
0.3	0.0000	0.0000	0.0043	0.0004	0.0338	0.1279
0.4	0.0000	0.0000	0.0034	0.0023	0.0430	0.1284
0.5	0.0000	0.0000	0.0016	0.0016	0.0219	0.0982
0.6	0.0000	0.0000	0.0016	0.0040	0.0136	0.0873
0.7	0.0007	0.0002	0.0120	0.0196	0.0148	0.0907
0.8	0.0001	0.0003	0.0051	0.0054	0.0088	0.0345
0.9	0.0080	0.0187	0.0030	0.0046	0.0119	0.0055
0.98	0.0417	0.0396	0.0161	0.0506	0.0470	0.0454

Note: p-values of Diebold-Mariano tests (HAR-RV-QRF vs. HAR-RV-QRF-RA forecasts) for three different forecast horizons. The HAR-RV-QRF model features 500 trees and a maximum of five terminal nodes. Null hypothesis: the series of forecasts are equally accurate. Alternative hypothesis: the forecasts from the HAR-RV-QRF model that features the risk aversion are more accurate. Loss function: check function of the *L*1 (absolute error loss) or the *L*2 (squared error loss) type. The models were estimated using a rolling-estimation window (2350 observations).

Table 9. Out-of-sample results (HAR-RV-ERF model).

Loss function	$h=1$	$h=5$	$h=22$
*L*1	0.0000	0.0330	0.2536
*L*2	0.0123	0.0302	0.0958

Note: p-values of Diebold-Mariano tests (HAR-RV-ERF vs. HAR-RV-ERF-RA forecasts) for three different forecast horizons. Null hypothesis: the series of forecasts are equally accurate. Alternative hypothesis: the forecasts from the HAR-RV-ERF model that features the risk aversion are more accurate. *L*1: absolute loss. *L*2: quadratic loss. The models were estimated using extreme random forests (500 trees, minimum node size was set to ten) and a rolling-estimation window (2350 observations).

Having obtained significant results for the predictive role of risk aversion, it is also interesting to compare how the HAR-RV-ERF model performs relative to the variants of the HAR-RV-RF model and the HAR-RV model. To this end, we construct the model confidence set (Hansen et al., 2011).[24] In total, we compare ten models: the HAR-RV-ERF-RA and HAR-RV-ERF models, the HAR-RV-RF-RA, and HAR-RV-RF models (using the three alternative calibrations given in Table 3), and the HAR-RV-RA and HAR-RV models. The results (details are not reported to save space) show that the HAR-RV-RF-RA and HAR-RV-ERF-RA models (both with a minimum node size of 10) are the top-ranked models for the short and the medium forecast horizon under *L*2 loss, while the HAR-RV models are the top-ranked models for the long forecast horizon. Under *L*1 loss, in turn, the HAR-RV-RF-RA and HAR-RV-ERF-RA models are top-ranked at the medium forecast horizon, while the HAR-RV-ERF-RA model is followed by the HAR-RV model at the short forecast horizon, and the HAR-RV-RF and HAR-RV-ERF-RA models lead the model confidence set at the long forecast horizon. Hence, the results show that, as one would have expected, the classic HAR-RV model is not a poor model per se. The results, however, also show that the classic HAR-RV model often does not reach the performance of random forests and the HAR-RV-ERF-RA model.

Table 10. HAR-RV-RF-RA/HAR-RV-ERF-RA model vs. HAR-RV-RA model.

Panel A: Statistical tests

Test	$h=1$	$h=5$	$h=22$
		HAR-RV-RF model	
DM	0.0794	0.0509	0.1583
CW	0.0015	0.0144	0.0163
		HAR-RV-ERF model	
DM	0.1158	0.0497	0.1176
CW	0.0028	0.0069	0.0121

Panel B: Economic tests

Model	Risk aversion		
	$\gamma=1$	$\gamma=3$	$\gamma=10$
HAR-RV-RF model	1.0465	1.8859	3.9239
HAR-RV-ERF model	−0.3860	0.9567	2.7636

Note: Panel A: p-values of Diebold-Mariano (DM) and Clark-West (CW) tests (HAR-RV-RF vs. HAR-RV-RF-RA and HAR-RV-RF vs. HAR-RV-ERF-RA forecasts) for three different forecast horizons. Null hypothesis: the series of forecasts are equally accurate. Results are for a quadratic loss function. The RF/ERF models were estimated using 500 trees and a minimum terminal node size of ten observations. All three models were estimated using a rolling-estimation window (2350 observations). Panel B: Difference (in percent) between the certainty equivalent returns (CER) that a forecaster attains upon using the HAR-RV-RF-RA or HAR-RV-ERF-RA models rather than the HAR-RV-RA model. A positive number indicates that a forecaster attains a higher CER ratio when using the HAR-RV-RF-RA or HAR-RV-ERF-RA models. Preference is of the constant-relative-risk-aversion (CRRA) type, where the parameter γ captures the degree of risk aversion. The risk-free interest rate is fixed at zero and there are no transaction costs. Portfolios rebalanced every day. Portfolio weights are restricted to the interval $[0, 1.5]$. The forecasting horizon is fixed at $h=1$. The random forests are estimated using the calibration also used in Panel A.

Building on these results, we compare in Table 10 the HAR-RV-ERF-RA and the HAR-RV-RF-RA models (minimum node size of 10) with the HAR-RV-RA model. In Panel A, we compare the models in statistical terms. We focus on the *L*2 loss function because under *L*2 loss the dominance of these two models is strongest according to our analysis of the model confidence set.[25] In line with the results of the model confidence set, we find that the HAR-RV-ERF-RA and the HAR-RV-RF-RA models dominate the classic HAR-RV-RA model for the two shorter forecast horizons, where the results of the Clark-West test are stronger at the long forecast horizon than those for the Diebold-Mariano test.

In Panel B, we compare the models in economic terms. To this end, we consider the decision problem of a forecaster who decides how to allocate funds across a riskless asset (we fix the riskless interest rate at zero) and an investment in oil. Like Cenesizoglu and Timmermann (2012), we simplify the decision problem by neglecting transaction costs. Furthermore, we assume that a forecaster optimizes his or her portfolio in every period of time, and we focus on the short forecast horizon. To compute the economic benefits of forecasts, we consider a forecaster whose utility function is of the constant relative risk aversion type ($U(W) = W^{1-\gamma}/(1-\gamma)$, where U = utility, W = wealth). We consider three levels of risk aversion ($\gamma = 3, 5, 10$; see Cenesizoglu & Timmermann, 2012). Finally, we calculate the

Table 11. Out-of-sample results (HAR-RV-RF model, alternative measure of *RV*).

Specification	$h=1$	$h=5$	$h=22$
Max. no. of terminal nodes = 5	0.0001	0.0387	0.0904
Max. no. of terminal nodes = 10	0.1109	0.0251	0.0827
Min. terminal node size = 10	0.0234	0.0264	0.1168

Note: p-values of Diebold-Mariano tests (HAR-RV-RF vs. HAR-RV-RF-RA forecasts) for three different forecast horizons. Null hypothesis: the series of forecasts are equally accurate. Alternative hypothesis: the forecasts from the HAR-RV-RF model that features the risk aversion are more accurate. Loss function: *L*2. The models were estimated using random forests consisting of 500 trees and a maximum of five terminal nodes.

Table 12. Out-of-sample results (HAR-RV-RF model, *h*-days-ahead forecasts).

Loss function	$h=1$	$h=5$	$h=22$
*L*1	0.0000	0.0000	0.0693
*L*2	0.0000	0.0130	0.1748

Note: p-values of Diebold-Mariano tests (HAR-RV-RF vs. HAR-RV-RF-RA forecasts) for three different forecast horizons. Null hypothesis: the series of forecasts are equally accurate. Alternative hypothesis: the forecasts from the HAR-RV-ERF model that features the risk aversion are more accurate. *L*1: absolute loss. *L*2: quadratic loss. The models were estimated using random forests (500 trees, the maximum number of terminal nodes was set to five), and a rolling-estimation window (2350 observations).

percentage difference between the certainty equivalent returns (CER) implied by the models. The results show that the HAR-RV-ERF-RA and the HAR-RV-RF-RA models yield higher CER than the HAR-RV-RA model, especially for the larger two risk-aversion parameters.

As a third extension, we repeat our analysis using an alternative measure of realized volatility in order to examine whether our results hold when we a use volatility estimator that is less robust to jumps. For this purpose, we use $RV^S = RV_t^B + RV_t^G$ as an estimator of realized volatility. Table 11 shows that all but two test results are significant.

As a fourth extension, we forecast *h*-days-ahead realized volatility rather than average realized volatility. Forecasting realized volatility *h*-days ahead is interesting because potential end-users of forecasts who trade, for example, in derivative contracts may be more interested in such *h*-days-ahead forecasts than in forecasts of average realized volatility. The results in Table 12 corroborate that risk aversion has predictive value for realized volatility.

Although not reported due to space considerations, we perform several additional extensions. 1) We study the predictive value of risk aversion for realized volatility at a monthly frequency and find that risk aversion does not improve forecast accuracy for monthly data, highlighting the benefits of using intraday data to study the predictive role of time varying risk aversion for volatility forecasting. 2) We study a model that features a monthly variance risk premium and observe insignificant results for this variable. 3) We consider a model that features the first difference in risk aversion and obtain significant test results. 4) Using the speculative ratio of Chan et al. (2015), we compare the model that features risk aversion with a model that features a measure of speculative activity in the oil market.[26] The results show that adding risk aversion to the predictors delivers significantly more accurate forecasts than adding the speculative ratio.

5. Concluding remarks

Using high-frequency data on WTI crude oil futures, we have examined the predictive power of time varying risk aversion for realized volatility in the crude oil market in the presence of various other covariates commonly utilized in earlier literature. Using random forests, we have found that risk aversion improves forecast accuracy for oil return volatility. We have argued that the predictive value of time-varying risk aversion over realized volatility is driven by the real-economic-activity channel and possible comovement of risk aversion with changes in economic conditions.

While the aim of our empirical research was not to run a horse race between different machine-learning techniques, it will be interesting to build on our analysis in future research to study the links between realized volatility of oil returns and risk aversion by means of other machine-learning techniques than random forests (e.g. Pradeepkumar & Ravi, 2017). Other measures of speculative activity could also be used to further rule out the speculation channel in the risk aversion-volatility link. Finally, the predictive role of behavioral factors over oil market volatility could be further explored via intraday trading data for oil traders.

Notes

1. Chatrath et al. (2015) and Phan et al. (2016) also forecast realized volatility of oil returns as derived from intraday data via regression and GARCH-based models instead of the HAR-RV framework. Asai et al. (2019, 2020) look at the role of jumps, leverage, spillovers, and geopolitical risks in forecasting the co-volatility of the oil and gold markets based on intraday data.
2. In recent research, Wen et al. (2019) report that economic market uncertainty (that is, the amount of risk) matters more for the realized volatility of oil-price returns than economic policy uncertainty.
3. Indeed, Qadan and Nama (2018) document evidence of a significant link between volatility in sentiment indices to volatility in oil prices, particularly after oil-based financial products started to attract greater attention from financial investors. Ye et al. (2020) document a (negative) impact of oil prices (and oil-price shocks) on investor sentiment (in the Chinese stock market) at different time frequencies (that is, short term versus long term).

4. Researchers often use the term volatility to denote the standard deviation of asset-price returns. To avoid any risk of confusion, we use the term realized volatility to denote the realized variance of oil-price returns and explicitly denote the square root of the realized volatility as the realized standard deviation of oil-price returns.
5. We also experimented with a rather long forecast horizon of 66 days. Results (for random forests with a maximum number of nodes set equal to five; p-values for the Diebold-Mariano test; absolute-error loss: 0.06; squared-error loss: 0.10) corroborated the predictive value of risk aversion for realized volatility. It should be noted that, for our out-of-sample analysis, we construct the data matrix such that the number of forecasts is the same for all forecast horizons.
6. For ease of notation, we do not use in Equation (4) an index for the forecast horizon, h and the time index, t.
7. This procedure requires a preset maximum number of terminal nodes or a preset minimum number of observations per terminal node.
8. Readers not familiar with regression trees may find it useful to consider a simple regression tree that features only one predictor variable and one split point. Such a regression tree resembles a conventional regression model with two dummy variables that capture whether the predictor variable takes on a value above or below the splitting point. When one starts growing a more complex regression tree, this amounts to replacing the two dummy variables with several additional dummy variables that partition the space of the predictor into finer and finer intervals. No empirical researcher would use such an increasingly complex model for volatility forecasting because it likely suffers from a severe overfitting problem. In order to overcome this problem, random forests combine a large number of (random) regression trees.
9. Jumps are associated with the discontinuous part of the price and can improve the overall fit of volatility models and help model and forecast realized volatility (see Corsi, 2008; Degiannakis & Filis, 2017; Duong & Swanson, 2015, among others). All of these studies employ HAR-type models with jump series. Jumps add a significant source of non-diversifiable locally sourced and difficult to predict volatility risk (Bollerslev et al., 2008). Both aggregate jump and volatility risk help explain the variation of expected returns (Cremers et al., 2015) and jumps improve forecasting of asset returns (Andersen et al., 2015), volatility (Duong & Swanson, 2015), the equity risk premium (Santa-Clara & Yan, 2010), and the variance risk premium (Li & Zinna, 2018).
10. For $q<2$, the jump variations are not always guaranteed to be finite.
11. In particular, they find that including $RVSJ_t$ in a HAR model significantly improves the forecast accuracy at the 1-day forecasting horizon, while including both $RVLJ_t$ and $RVSJ_t$ achieves significantly higher forecast accuracy at longer forecasting horizons (e.g. weekly or monthly).
12. Liu et al. (2015) note that 5-minute is a sampling frequency adequate for liquid assets like WTI futures. Thus, the sampling frequency used in our analysis is not too low to lead to poor data analysis and it is not too high to induce spurious jumps due to market frictions.
13. The models are estimated by the ordinary-least-squares technique. All estimation results documented in this research were computed using the R programming environment (R Core Team, 2019). The Newey-West robust standard errors reported in Table 1 were computed using the R packages "sandwich" (Zeileis, 2004).
14. We report p-values computed using the R package "forecast" (Hyndman, 2017; Hyndman & Khandakar, 2008).
15. We also studied scaled forecast errors computed by dividing the forecast error by actual realized volatility. The results (not reported) for scaled forecast errors turned out to be in the majority of cases qualitatively similar to the results we report in this paper. Scaled forecast errors account for heteroskedasticity (for such a heteroskedasticity adjustment, see e.g. Bollerslev & Ghysels, 1996). At the same time, scaling gives rise to a skewed distribution due to small realizations of realized volatility. See Hyndman and Koehler (2006).
16. Visual inspection of partial dependence functions (not reported due to space considerations) confirmed that the link between risk aversion and realized volatility is nonlinear, where higher values of risk aversion are associated with higher values of realized volatility, a pattern one would expect given that both realized volatility and risk aversion tend to be higher in times of financial stress than during tranquil times. Importantly, the partial dependence functions exhibit a moderate slope for several lower decentiles of risk aversion, while the slope substantially rises for the larger decentiles.
17. We use the R package "randomForest" to estimate random forests (see Liaw & Wiener, 2002). We set the number of trees to 500 and we describe #floor(total number of covariates/3) random covariates for splitting. We use the same to estimate the models described in Section 4.3.
18. Relative importance is measured by inspecting the increase in node purity (as measured by the residual sum of squares across all trees) that results from the inclusion of a variable to the list of predictors. We report the mean rank of a covariate across all rolling-estimation windows.
19. The low ranks of the jump covariates is not unexpected because we study a jump-robust estimator of realized volatility.
20. The pseudo-R-squared uses the so-called out-of-bag data (that is, the quasi-out-of-sample data not used for tree building when bootstrapping a random forest) to assess model performance (for further details, see Liaw & Wiener, 2002).
21. In a recent application of quantile random forests to stock-returns volatility forecasting, Gupta et al. (2019) provide a compact description of the technical details of quantile random forests. We use the R package "quantregForest" (Meinshausen, 2016) to estimate the HAR-RV-QRF model.
22. On the role of asymmetry of a forecaster's loss function for forecasts of crude oil-returns volatility, see Gkillas et al. (2020).
23. We use the R package "extraTrees" to estimate random forests (see Simm et al., 2014).

24. We use the R add-on package "MCS" to construct the model confidence set. See Bernardi and Catania (2014) for details.
25. It also should be noted that, as shown by Equation (4), the *L2* loss function underlies the search-and-partition algorithm that forms the foundation of random forests.
26. The speculative ratio is computed as the trading volume divided by the open interest for crude oil futures (data obtained from Commodity Systems Inc.).

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